

Measuring total causal effects of prompting strategies on LLM behaviour

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1 Introduction

Large Language Models (LLMs) have become increasingly widespread in multilingual and multicultural contexts. However, their performance varies significantly depending on the language and cultural background of the prompt.

This work investigates the causal effect of different prompting strategies on the ability of a multilingual LLM to retrieve culturally specific knowledge. Specifically, three prompting strategies are compared to assess whether querying a model in the native language of the target culture or assigning it a cultural persona improves the performance on culturally specific everyday knowledge questions with respect to a neutral English prompt. The results show that prompting the LLM in the local language improves its ability to retrieve culturally specific knowledge. Whereas assigning a cultural persona has little to no effect on the behavior of the model.

1.1 Literature

Several works have addressed the evaluation of LLMs across languages and cultures.

BLEnD [1] is a benchmark for the evaluation of everyday knowledge across diverse cultures and languages. It studied 16 different cultures and 13 different languages by querying multiple LLMs with different characteristics and compared their answers to those given by human annotators. Their results show that LLMs yield better cultural knowledge accuracy on cultures that are highly represented online; mid-to-high-resource cultures achieve a better performance with prompts in their local language, while for low-resource languages, the LLMs perform better in English.

The application of causal inference to NLP frameworks is investigated in [2]. Causality is rarely investigated in NLP, because of the challenges that arise from trying to estimate effects in the textual domain.

Multilingual LLMs have advanced increasingly in the past years, the authors of Tiny Aya [3] propose a lightweight 3.35B-parameter pretrained model that is able to achieve a strong and balanced performance across 70 languages. Different versions of Tiny Aya are presented, each focused on a specific region of the world. Tiny Aya

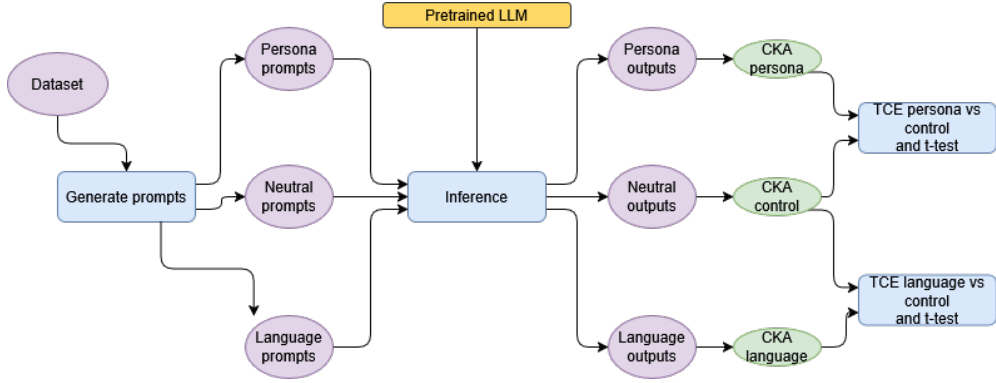


Fig. 1 Pipeline of the project

represents a recent compact model that guarantees efficiency, practical deployment and balance. It also proves effective as an alternative to brute force scaling.

2 Research question and methodology

This work investigates whether and to what extent the formulation of a prompt influences the ability of a large language model to retrieve culturally specific knowledge. Specifically, the question is: does assigning a cultural persona to the model, or prompting it in the native language of a culture, causally improve its performance on culture-specific factual questions compared to a neutral English prompt?

More formally, let M be a multilingual LLM and Q a set of culturally grounded questions from the BLEnD benchmark. For each question three prompt formulations are defined: a control prompt (neutral English), a persona prompt (English with an explicit cultural role assignment), and a language prompt (native language of the target culture). The outcome variable is the Cultural Knowledge Accuracy (CKA), defined in section 3.2.

Each prompting strategy is modeled as a treatment and its Total Causal Effect (TCE) on CKA is estimated with respect to the control group, as defined in section 3.2.

2.1 Pipeline

The project follows a rigorous pipeline to achieve the expected output.

1. Dataset selection
2. Prompt generation
3. LLM inference
4. Output scoring
5. Statistical analysis

Figure 1 shows a flowchart of the pipeline, which is also described in the following.

First, a multilingual dataset of culturally grounded questions (the BLEnD benchmark, 3.1) was chosen and analyzed.

Second, different versions of the same prompt were formulated in order to measure the causal effect of three different prompting strategies for the same question: one prompt in plain English, one prompt in the local language of the country it referred to, and one prompt that explicitly assigned a cultural persona to the LLM.

Third, the generated prompts were used to perform inference using an LLM as described in the following section, which, based on these prompts, provided an output in the form of a short answer (possibly a single word or a few words at most).

Fourth, the answers of the LLM were collected and scored according to a binary score, based on their accuracy. The average of these scores represents the target outcome: the Cultural Knowledge Accuracy of the model.

Finally, the statistical significance of applying the “treatment” (language and persona prompts) was computed.

All of these steps and their implementation decisions are further discussed in section 3.3.

2.2 Model

The LLM for this project was selected according to two main criteria: it needed to be a multilingual model, including non-Western languages, and it had to be light enough to be run on limited resources.

For this reason, Cohere Labs’ *Tiny Aya Global* was chosen [3]. Tiny Aya Global is an open-weight pretrained model with 3.35 billion parameters optimized for multilingual representations: it comprises more than 70 languages, including low-resourced ones. It is specifically designed for deployment under realistic compute constraints.

Unlike other larger models, such as Aya Expanse 8B, which was also considered, Tiny Aya Global fits comfortably within the constraints of a free-tier GPU (T4, provided by Google Colab), while maintaining strong multilingual capabilities.

3 Experimental results

3.1 Dataset

The dataset used for this project is the BLEnD Dataset [1], a benchmark to “evaluate LLM’s everyday knowledge across diverse cultures and languages”. The BLEnD dataset includes two different types of question-answer pairs: short answer questions and multiple choice questions. It includes questions from 16 different countries in 13 languages. The questions are provided both in English and in the local language of the country they are referring to. They focus on everyday cultural knowledge, such as information about food, school, or leisure with respect to a specific country.

Alongside the questions, the authors of the dataset provided annotations, i.e. answers to the short answer questions format given by annotators from each country.

For this project, only the short answer questions were used to produce the desired results, as they were already available in English and in the local language, for both

questions and answers, while the multiple choice questions only had an English version. A subset of countries was chosen, including Algeria, China, Greece, Indonesia, and Spain, in order to represent different kinds of languages with different alphabet writing systems, some of which are relatively underrepresented in LLM training corpora, whereas others are comparatively well represented. The choice of the language was made also keeping into account the languages supported by the model and ensuring that each language was represented only once (e.g., Spain and Mexico). English-speaking countries were discarded because they defeated the purpose of generating a language prompt.

The final experimental dataset consisted of 500 questions for each of the five selected countries, for a total of 2500 questions.

3.2 Metrics

Cultural Knowledge Accuracy (CKA) is defined as the proportion of correctly answered questions. Each model output receives a binary score: 1 if it contains at least one valid reference answer from the BLENd dataset, and 0 otherwise. CKA is then computed as the average of these binary scores over a set of questions.

Since the aim of the project is to detect whether the LLM responds differently to diverse prompting strategies, the main metric used to evaluate it is the Total Causal Effect (TCE). The TCE, equivalent to the Average Treatment Effect (ATE) [2] in this experimental setting, is computed as:

$$TCE = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] \quad (1)$$

where $Y(0)$ is the control group and $Y(1)$ is the treatment group, in this case persona prompt and language prompt.

Finally, statistical significance measures, i.e. t-tests, were employed to verify if the effect of the treatment is statistically relevant with respect to the non-treated prompting strategy. These statistical tests produced t-statistics and p-values, which will be discussed in section 3.4.

3.3 Experimental setup

All experiments were run on Google Colab using the free T4 GPU with Python 3.

As stated in section 3.1, only a subset of the BLENd dataset was employed in this project: in particular the *short answer questions* format from Algeria, China, Greece, Indonesia, and Spain. These questions were collected from the dataset files and used to generate three different kinds of prompts:

- Neutral (or control) prompts, reporting the question as is written in the dataset
- Persona-based prompts, adding the statement “As a person from [country],” at the beginning of each question in the dataset
- Local-language prompts, taking the translation of the neutral question, already provided by the authors of the BLENd dataset

The sentence “Provide a single answer without any explanations.” was then attached to each prompt, in order to get an answer in a format similar to that of the valid answers provided by the dataset.

Inference was carried out with the Tiny Aya Global model, whose specifics are described in section 2.2. Inference was implemented via a HuggingFace¹ pipeline object, which offers an API dedicated specifically to text generation for Tiny Aya Global, among others. The LLM generated its outputs based on specific parameters:

- `max_new_tokens = 50`, which is the upper bound on the length of the generated text
- `temperature = 0.1`, which makes the outputs more predictable
- `do_sample = True`, which ensures some randomness in the results; however, it is bounded by the low value of the temperature

Choosing a low temperature value makes the model more heavily biased towards the most statistically probable next tokens, but ensures reproducibility of results. This is preferable in the case of this work, since the task aims for accuracy rather than creativity.

Before evaluation, both the model outputs and the reference answers were converted to lowercase and stripped of punctuation. An output was considered correct if it contained at least one of the valid answers provided by the BLEND dataset. Then CKA was computed for each group of prompts (control, persona, language) for all the selected countries, as described in section 3.2.

Then, the TCE (Total Causal Effect) was computed by subtracting the CKA (essentially the mean of the scores) of the *control* group from the CKA of the *treatment* group (either persona or language).

Finally, statistical significance of the results was computed via an independent samples t-test on both treatment groups for all the countries analyzed. A result was deemed to be statistically significant if its `p_value` was lower than or equal to a value $\alpha = 0.05$. Welch’s t-test was chosen over students’ t-test because it does not assume equal variances in the samples and is therefore more robust when comparing groups with potentially unequal variances.

The implementation of this project can be found on GitHub at the following link: <https://github.com/alicee127/NLP-project>.

3.4 Results

Figure 2 shows the values of the CKA metric for the three different prompting strategies divided by country. The barplot shows that language prompts achieve a higher accuracy for all languages except for Arabic (Algeria) and Greek, while persona-based prompts achieve similar CKA results to neutral prompts for all countries.

Figure 3 shows the total causal effect (TCE) of the prompting strategy with respect to the control group for each country: the effect of formulating a prompt in the local language appears to be far greater than the effect of the persona-based prompting strategy. For Greece and Algeria, both prompting strategies seemed to yield worse performance than the control.

¹<https://huggingface.co/>

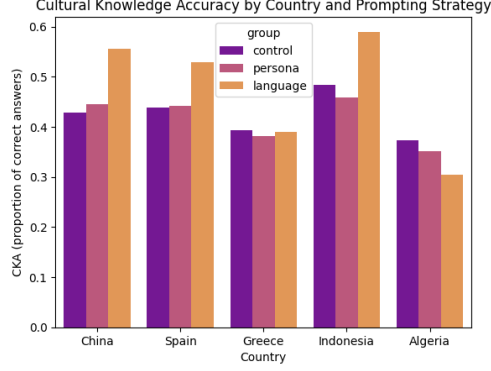


Fig. 2 Cultural Knowledge Accuracy for different prompting strategies across countries

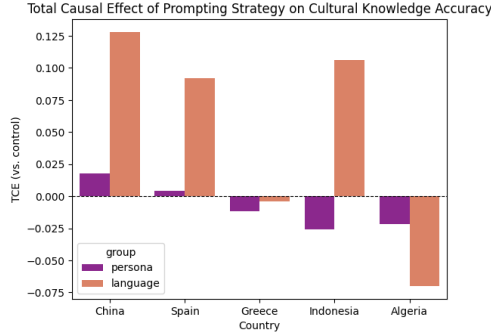


Fig. 3 Total Causal Effect (or Average Treatment Effect) for each country for persona-based prompts and local language prompts with respect to neutral prompts

Table 1 shows the values of CKA and TCE for each prompting strategy across all countries.

Table 2 shows the results of the t-test: language prompts appear to be statistically significant for all languages except Greek, while persona prompts are not deemed as statistically significant for any country.

4 Conclusion

This work investigated the causal effect of different prompting strategies on the cultural knowledge accuracy of a multilingual LLM. The experiments showed that formulating a prompt in the native language of the country improves the quality of the model’s answers for most of the analyzed countries. This suggests that the model probably learned cultural representations connected to language, not only to the semantic content of the questions.

The fact that persona-based prompts do not have a statistically significant effect on any culture suggests that adding an explicit cultural context to an English prompt does not effectively allow the LLM to collect specific cultural knowledge.

Country	Group	CKA	TCE
China	Control	0.428	-
	Persona	0.446	0.018
	Language	0.556	0.128
Spain	Control	0.438	-
	Persona	0.442	0.004
	Language	0.53	0.092
Greece	Control	0.394	-
	Persona	0.382	-0.012
	Language	0.39	-0.004
Indonesia	Control	0.484	-
	Persona	0.458	-0.025
	Language	0.59	0.106
Algeria	Control	0.374	-
	Persona	0.352	-0.022
	Language	0.304	-0.07

Table 1 This table shows the values of CKA and TCE for each prompting strategy and each country

Country	Group	t-val	p-val	Significant
China	Persona	0.573	0.567	False
	Language	4.078	0.000049	True
Spain	Persona	0.127	0.899	False
	Language	2.920	0.00358	True
Greece	Persona	-0.389	0.697	False
	Language	-0.129	0.897	False
Indonesia	Persona	-0.823	0.411	False
	Language	3.377	0.000761	True
Algeria	Persona	-0.723	0.470	False
	Language	-2.342	0.0194	True

Table 2 This table displays the results of Welch’s independent samples t-tests for each country and prompting strategy

4.1 Limitations

Despite the promising results, this work is subject to some limitations.

In particular, the selected LLM is a small model, which, despite being more flexible in deployment, might limit the accuracy of the inference.

Furthermore, no tokenization nor stemming of the answers of the model was performed. This, along with partial matching of the outputs and the valid reference answers, might underestimate or overestimate the value of CKA.

The selected subset of countries is not representative of a wide enough range of languages of the world.

The cultural persona assignment was implemented using a fixed template. Different persona prompts formulations might yield different results.

Finally, the low temperature value assigned to the model limits the variability of its answers.

4.2 Future work

The limitations identified above suggest some directions for future work, such as:

- Comparing the performances of multiple models, similarly to what was done in [1]
- Testing the same pipeline with a different model or with another version of Cohere Labs’ Tiny Aya, e.g. Tiny Aya core, Tiny Aya Earth (languages across Africa and West Asia), Tiny Aya Water (South Asian languages), and Tiny Aya Fire (languages of Asia-Pacific and Europe regions)
- Implementing stemming/lemmatization/tokenization, like in [1]
- Extending the analysis to other, possibly lower resource, cultures

AI usage statement

Generative AI (Claude 4.6) was used in this project in a limited capacity for assistance in debugging code and organizing ideas. All the project’s implementation decisions, the design of the experiment, analysis and writing of the paper are my own work.

References

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